

Mohak Kapoor

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SUMMARY

I build AI systems end to end, from the model logic down to the infrastructure that serves it. At HumanizeIQ I designed production MCP servers and wired them into an orchestrator for agent-driven tasks, and on my own I built a multi-repo RAG assistant with a custom hybrid retrieval setup and a framework just to evaluate how well it actually performs. I'm comfortable owning a project from FastAPI backend to Docker deployment. I like to dig into how things work under the hood, and I'm looking for a team where I can take on real ownership and learn.

PROJECTS

CodeAtlas : Multi-Repo Agentic RAG Assistant | LangChain, Qdrant, Gemini (Live Link, GitHub Link) Present

- Built an **agentic RAG engine** aggregating an entire GitHub portfolio by topic rather than a single repository, solving the "README domination" problem via a **hybrid retrieval pipeline** (dense + BM25 with cross-encoder reranking) that lifted context recall **0.48** → **0.83** over a naive baseline on a 52-case benchmark.
- Implemented **structure-aware semantic chunking** via Python **AST parsing** (functions/classes as discrete units) paired with Markdown header-based splitting; synced incrementally via **manifest.json**-tracked commit diffs to avoid full re-indexing.
- Designed **JudgeKit**, a supplementary dual-agent **LLM-as-judge eval framework** (Llama 3.3 70B) scoring retrieval (**Precision: 0.78, Recall: 0.83**) and generation (**Faithfulness: 0.88, Relevance: 0.91, Correctness: 0.82**) quality, used to benchmark three retrieval architecture iterations.

Network IDS Detection Engine | Autoencoder, PyTorch, FastAPI, Docker (Live Link, GitHub Link) Jan–May 2026

- Engineered a **two-stage hybrid IDS** on 2.8M rows of CICIDS2017. Unsupervised anomaly detection flags candidates for supervised classification, mirroring production platforms like Snort and Darktrace; built **end-to-end data pipeline** handling ingestion, EDA, preprocessing, and temporal train/test splitting to prevent distribution leakage across unseen attack types.
- Hybrid Model 92.4% accuracy** across four benchmarked architectures (XGBoost (F1: 0.97), PyTorch FFNN (F1: 0.98), SVM (F1: 0.96)); Denoising Autoencoder (AUC: 0.7801) outperformed Isolation Forest on zero-day detection, with IncrementalPCA **cutting feature space 51%** (98.97% variance retained) .
- Shipped a Dockerized FastAPI inference engine for the hybrid IDS model to edge hardware (Raspberry Pi 5), exposing predictions via a public endpoint secured with Bearer token auth over Cloudflare Tunnel, with a WebSocket dashboard streaming live latency metrics.

EXPERIENCE

HumanizeIQ

Jun 2025 – Oct 2025

AI Intern

Hazlet, New Jersey (Remote)

- Designed and built an n8n-orchestrated LLM evaluation pipeline** for scoring recruiter call performance, including the **evaluation logic itself**: script-adherence scoring against transcript, recruiter identity, and job role.
- Restructured an existing Postgres call-transcript database** into normalized **calls** and **follow-back** tables, then built a **FastAPI backend** (GET endpoints for transcript, identity, and script retrieval; POST endpoints to persist scores) called end-to-end by the evaluation pipeline, plus a second backend serving report data to the client-facing dashboard, cutting analyst turnaround from 2 hours to under 5 minutes.
- Designed two production MCP servers in TypeScript on Cloudflare Workers**: an image-generation MCP calling an internal LLM gateway with outputs persisted to Cloudflare R2, and a document-orchestration MCP composing multi-modal HTML reports (text, maps, generated images) by invoking the first MCP as a tool.
- Led daily Scrum stand-ups** in an agile environment for the full internship; independently built the **AnythingLLM orchestrator integration** for both MCPs, enabling agent-driven document and image generation tasks.

CERTIFICATIONS & ACHIEVEMENTS

- JPMorgan Chase & Co. - Quantitative Research Job Simulation**, Forage (Certificate)
- Machine Learning Specialization**, DeepLearning.AI, Stanford University (Certificate)
- 125th Rank, LeetCode Biweekly Contest 184
- Winner, OSDHack'23 (JIIT)

TECHNICAL SKILLS

Languages	Python (Advanced), TypeScript, C++ (Basics)
LLMs & GenAI	LLM Integration (GPT-4, Gemini), MCP, RAG, Agentic Systems, LangChain, n8n, AnythingLLM, Agentic Tool Calling
Systems & Platform	Docker, Linux, Cloudflare Workers, Cloudflare Tunnel, Edge Deployment, OOP, Agile/Scrum, SDLC
MLOps & Deployment	FastAPI, Flask, REST APIs, CI/CD, Experiment Tracking, Model Monitoring
Data & Retrieval	Qdrant, Vector Databases, HuggingFace Embeddings, Hybrid Retrieval, Retrieval Evaluation
Frameworks & Data	PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, PostgreSQL, SQL, Git
ML & Modeling	Supervised Learning, XGBoost, Time-Series Forecasting, Anomaly Detection, Feature Engineering, DSA

EDUCATION

Jaypee Institute of Information Technology

Noida, India

B.Tech in Electronics and Communication Engineering

2022-2026

- Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Computer Systems, Queueing Theory